TrashUQ

Edge AI + Federated Learning Platform

Aleix Bertran · Aleix Rosinach
Bru Pallàs · Pol Llinàs

May 2026 · Real-Time Monitoring & Scaling Validation

Project Overview

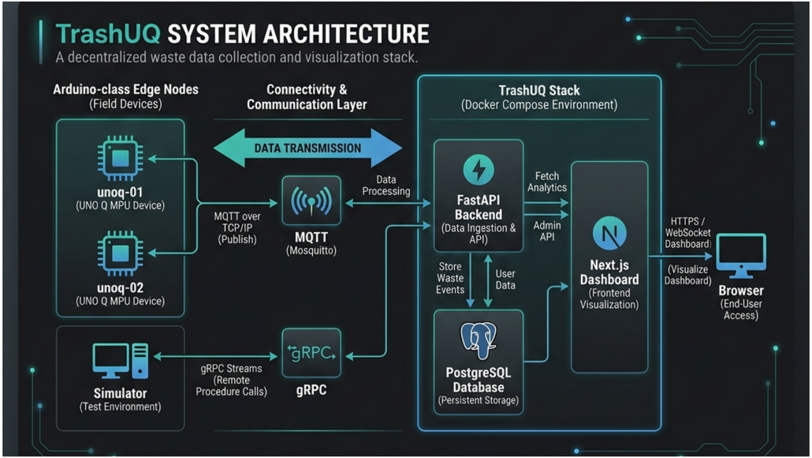
TrashUQ is an end-to-end platform for intelligent waste monitoring.

- **Decentralized Training**: Federated Learning at the Edge.
- **Real-Time Pipeline**: MQTT telemetry & WebSocket dashboard.
- **Robust Persistence**: PostgreSQL storage of every event.
- **Validated Hardware**: Deployed on Arduino-class MPU nodes.

Core Stack

- **Backend**: FastAPI & gRPC
- **Edge**: TFLite & Python
- **Dashboard**: Next.js
- **Broker**: Mosquitto

System Architecture



Three-Stage Pipeline: Edge Devices → Connectivity → Deployable Stack

Edge Node Intelligence

Device Layer

- **Hardware**: UNO Q MPU (Arduino-class).
- **Runtime**: TFLite-based material classification.
- **Simulator**: Full-fidelity reproducible demo mode.

Classification Path

- Cardboard
- Glass
- Paper
- Plastic

Telemetery: Published via MQTT topics.

The Communication Backbone

MQTT Telemetry

- Real-time status, metrics, and classification streams.
- Low-bandwidth, high-reliability pub/sub.

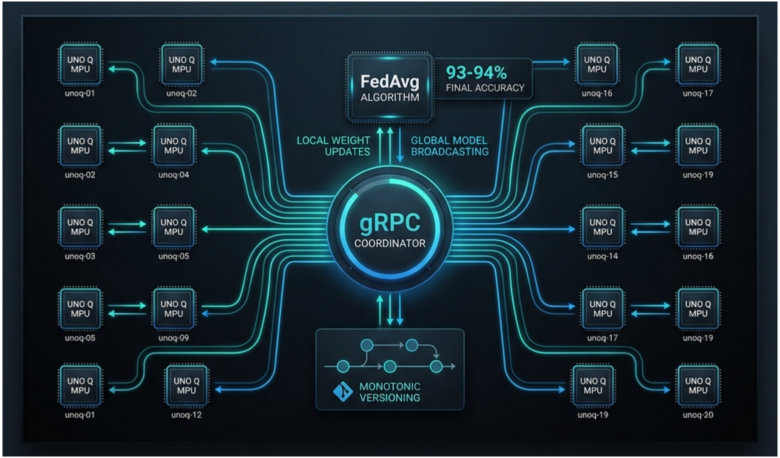
gRPC Coordination

- High-performance binary RPC for model weight transfer.
- Concurrency-safe: Rejects stale updates during rounds.

Data Flow

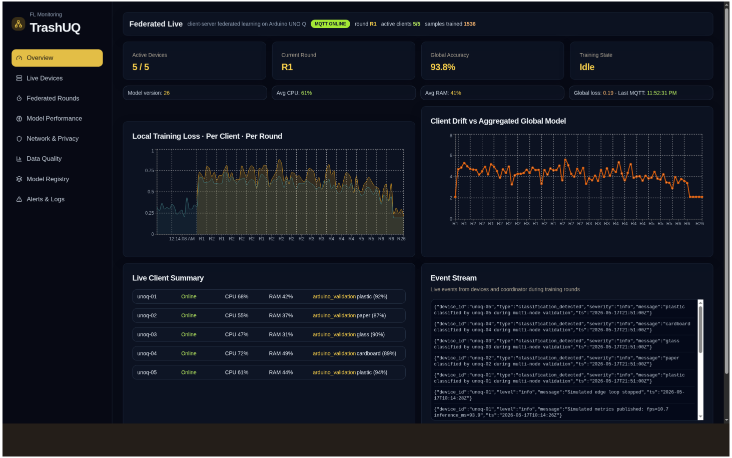
`arduino/+/#` → Backend → PostgreSQL → Dashboard

Federated Learning Scaling



FedAvg Validation: Scalability study across 2, 5, 10, and 20 clients.

Next.js Live Dashboard



Bootstrap + MQTT WebSockets: Real-time charts and device summaries.

Part A: Deployed Validation

- **Infrastructure**: 2 nodes (`unoq-01`, `unoq-02`) on `bepes-server`.
- **Persistence**: 282 MQTT messages stored with full fidelity.
- **Monotonicity**: Advanced `model_version` from 0 to 26 over 26 rounds.
- **Reliability**: 0% invalid payloads, 0% dropped messages.

Node	Messages	Mean Accuracy
unoq-01	147	92.1%
unoq-02	135	87.1%

Part B: Scaling Performance

- **Experiment**: FedAvg scaling study (2 to 20 clients).
- **Condition**: Non-IID Dirichlet partitioning ($\alpha = 0.3$).
- **Convergence**: Stable convergence with **93-94% final accuracy**.
- **Efficiency**: predictable linear growth in communication cost.

Conclusion: System maintains high performance at scale under heterogeneous data.

Summary & Team

TrashUQ Success

- Full edge-to-cloud integration.
- Validated FL coordinator.
- Real-time dashboard.

The Team

- Aleix Bertran
- Aleix Rosinach
- Bru Pallàs
- Pol Llinàs

Thank You!